

3.1.4 Competitive and Concentrated Markets

Market Structure: Overview

Market structure describes the competitive environment in which firms operate. It is determined by factors like the number of firms, barriers to entry, product differentiation, and the degree of market power. The four main structures form a spectrum from most competitive to least competitive:

Feature	Perfect Competition	Monopolistic Competition	Oligopoly	Monopoly
Number of firms	Very many	Many	Few dominant firms	One
Barriers to entry	None	Low	High	Very high / impossible
Product type	Homogeneous (identical)	Differentiated	Differentiated or homogeneous	Unique (no close substitutes)
Price-setting power	None (price taker)	Some (limited by substitutes)	Significant (interdependent)	Considerable (price maker)
Supernormal profit in LR	No	No	Yes (possible)	Yes
Key feature	Price-taking	Product differentiation	Interdependence	Barriers to entry

Real-world examples	Currency markets, commodity farming	Restaurants, hairdressers, coffee shops	Supermarkets, airlines, mobile networks, petrol	Thames Water, Network Rail (regulated)
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Exam Tip: In reality, very few markets fit perfectly into one category. Most real markets fall somewhere between these theoretical extremes. When analysing a market in an exam, identify its key features and discuss which model it most closely resembles, while noting deviations.

Perfect Competition

Perfect competition is a theoretical market structure characterised by many small firms selling identical products, with perfect information, no barriers to entry or exit, and where no individual firm can influence the market price.

Assumptions of Perfect Competition

All of the following must hold simultaneously:

1. **Many buyers and many sellers** — each is so small relative to the market that no single buyer or seller can influence the price
2. **Homogeneous (identical) products** — consumers see no difference between the output of different firms
3. **Perfect knowledge / information** — all buyers and sellers have complete information about prices, quality, and production methods
4. **No barriers to entry or exit** — firms can freely enter or leave the industry at no cost
5. **Firms are price takers** — they must accept the market price determined by supply and demand
6. **Perfectly mobile factors of production** — resources can move freely between uses
7. **No externalities or government intervention**

Because of these assumptions, perfect competition is largely a theoretical benchmark rather than a real market. However, some markets approximate it — agricultural commodity markets (wheat, milk),

foreign exchange markets, and some online markets where products are standardised and comparison is easy.

The Firm's Demand Curve in Perfect Competition

Because the firm is a price taker, it can sell any quantity at the market price. Its demand curve is **perfectly elastic (horizontal)** at the market price. This means:

$$AR = MR = \text{Price (all are constant and equal)}$$

Short-Run Equilibrium: Supernormal Profit

In the short run, a perfectly competitive firm can earn supernormal profit if the market price exceeds average cost at the profit-maximising output ($MC = MR$).

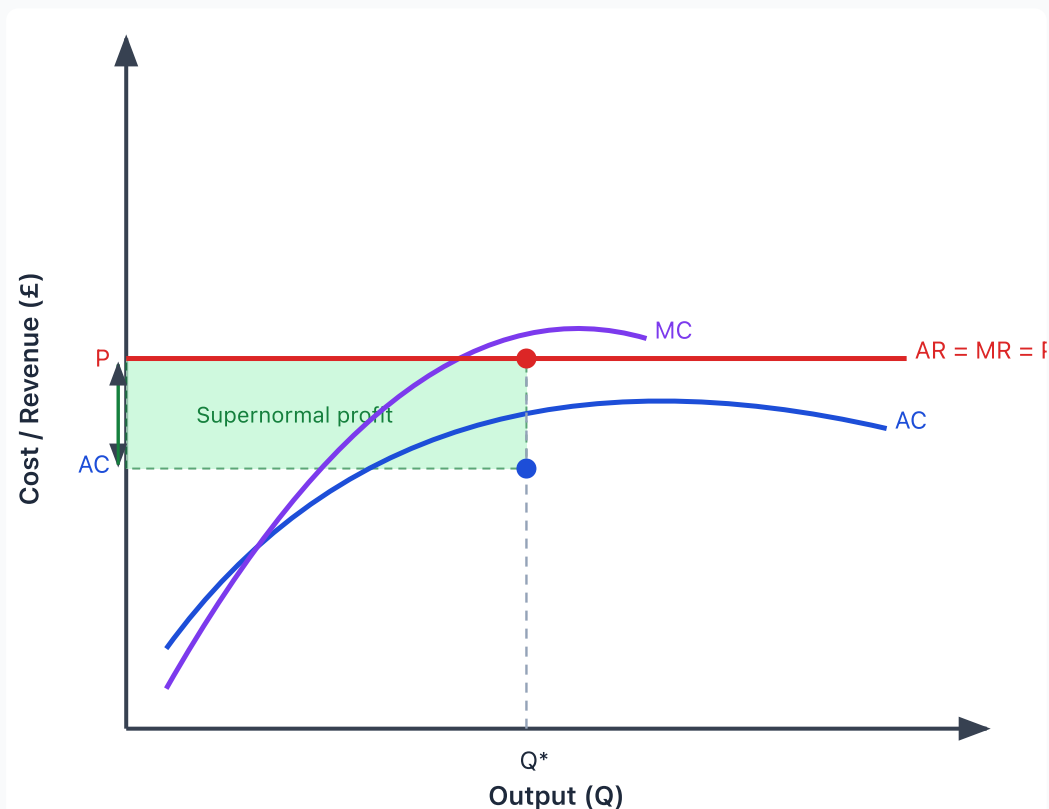


Figure 1: A perfectly competitive firm earning supernormal profit in the short run ($P > AC$ at $MC = MR$)

Long-Run Equilibrium: Normal Profit Only

Supernormal profits cannot persist in the long run because there are **no barriers to entry**:

1. Existing firms earn supernormal profit ($P > AC$)
2. This attracts new firms into the market
3. Market supply shifts right → market price falls
4. This continues until $P = AC$ (minimum) and all supernormal profit is competed away
5. Firms earn only **normal profit**

The reverse process works for losses: if firms make subnormal profit ($P < AC$), firms exit → supply shifts left → price rises → losses eliminated.

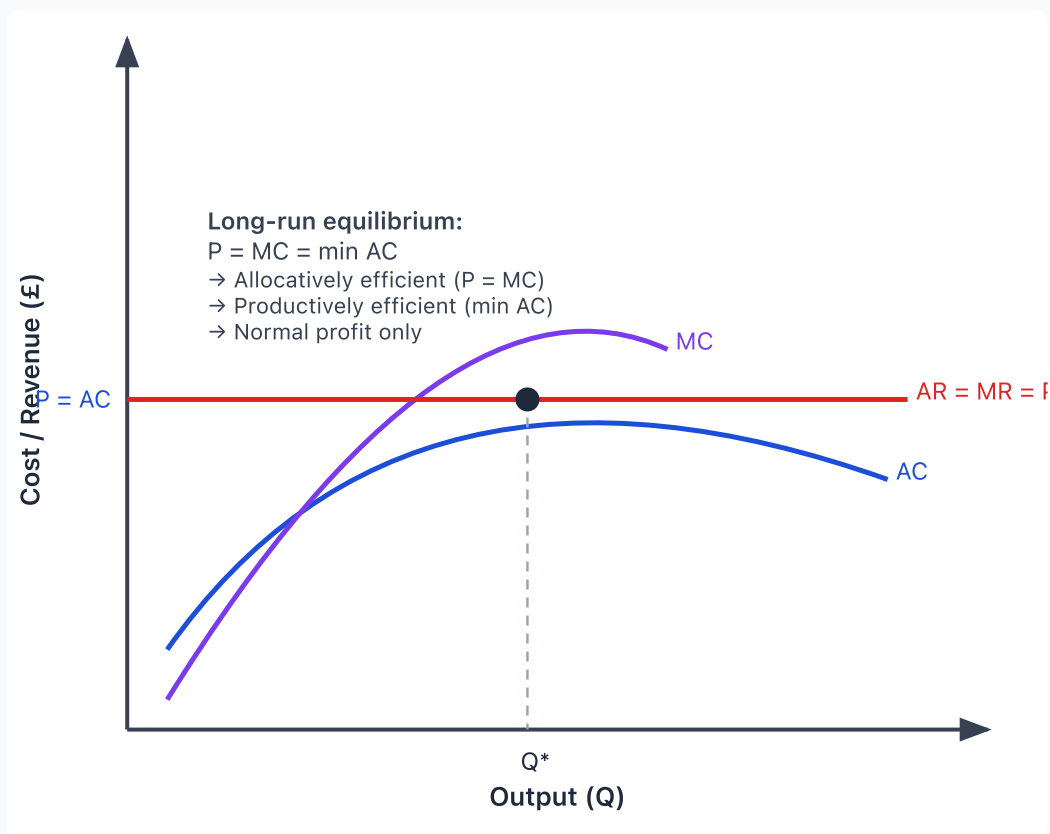


Figure 2: Long-run equilibrium in perfect competition — $P = MC = AC$ (minimum). Firms earn normal profit only. Both allocative and productive efficiency are achieved.

Long-Run Equilibrium Conditions

$$P = MC = AC \text{ (minimum)} = AR = MR$$

This means:

- **Allocative efficiency** ($P = MC$) — the value consumers place on the last unit equals the cost of producing it
- **Productive efficiency** (production at minimum AC) — output is produced at the lowest possible cost per unit
- **Normal profit only** — no wasteful supernormal profit, but enough to keep firms in the industry

Real-World Approximation: UK Dairy Farming

UK dairy farming approximates some features of perfect competition: many small producers, a largely homogeneous product (milk), relatively low barriers to entry/exit, and farmers are essentially price takers — they accept whatever the processors (Arla, Muller) and supermarkets offer. In 2015–16, the farm-gate milk price fell below the cost of production (about 25p/litre vs production costs of 28p/litre). Many farmers made losses and exited — over 1,000 dairy farms closed in two years. This is exactly what the model predicts: losses → exit → supply falls → price eventually recovers.

Monopoly

Pure monopoly — a market with a single seller providing a product with no close substitutes. In practice, pure monopoly is extremely rare.

Legal/working monopoly — in UK competition law, a firm is classified as having monopoly power if it holds **25% or more** of the market share. This is the more practical definition used by regulators (CMA in the UK).

Sources of Monopoly Power: Barriers to Entry

Monopoly power depends on barriers that prevent new firms from entering the market and competing away supernormal profits:

Barrier Type	How It Works	Real-World Example
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Economies of scale / Natural monopoly	One firm can supply the entire market at lower cost than two or more firms. The MES is so large relative to demand that only one firm is viable.	Water supply (Thames Water), electricity transmission (National Grid), railways (Network Rail)
Legal barriers — Patents	Patents give a firm exclusive rights to produce a product or use a process for a set period (usually 20 years), preventing imitation.	Pharmaceutical companies hold patents on new drugs — Pfizer's patent on Viagra lasted until 2013, allowing it to charge monopoly prices
Legal barriers — Licences	Government-granted licences restrict who can operate in a market.	TV broadcasting licences, taxi licences (London black cabs), spectrum licences for mobile networks
Brand loyalty & advertising	Massive spending on branding creates a perception of superiority that new entrants cannot easily match.	Coca-Cola's brand, estimated at \$90+ billion, makes it almost impossible for a new cola to compete seriously
Control of essential resources	Owning a key input that competitors need.	De Beers historically controlled 80%+ of global diamond supply
Network effects	The product becomes more valuable as more people use it, making it extremely hard for a new competitor to attract users.	Facebook/Meta — a new social network is worthless if nobody else is on it. WhatsApp has 2+ billion users — switching costs are enormous.
Sunk costs	Large irrecoverable investments required to enter a market deter potential entrants — if the entry fails, those costs are lost.	Building a new car factory costs billions. If the brand fails, the factory cannot easily be converted.

Predatory pricing

Existing monopolist temporarily cuts prices below cost to drive out a new entrant, then raises prices again. Illegal but hard to prove.

Allegations against Amazon for selling goods below cost to eliminate smaller competitors

Monopoly Profit Maximisation Diagram

A monopolist faces a downward-sloping demand curve (it IS the market). It maximises profit where $MC = MR$, then charges the price consumers are willing to pay from the demand (AR) curve at that output level.

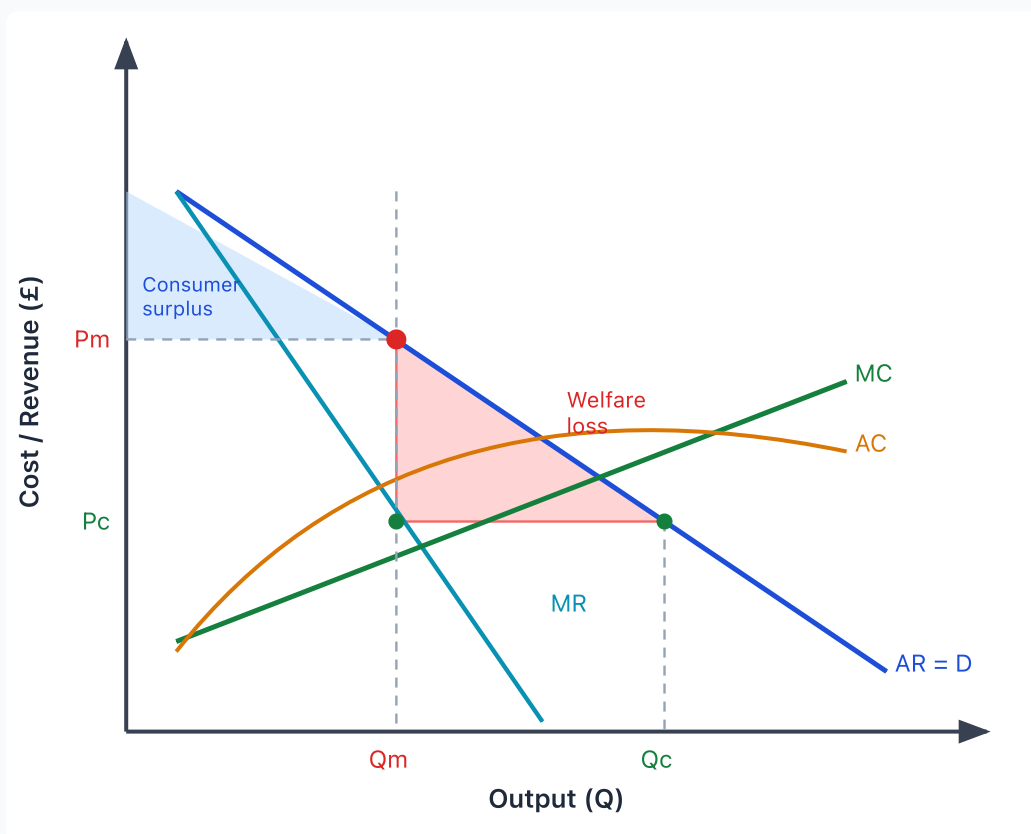


Figure 3: Monopolist profit maximisation at $MC = MR$. Price P_m is above AC , generating supernormal profit. The deadweight loss (DWL) triangle shows the welfare loss compared to competitive output Q_c .

Monopoly vs Perfect Competition: The Key Comparison

Compared to a perfectly competitive market with the same cost structure, a monopolist:

- **Charges a higher price** ($P_m > P_c$)
- **Produces less output** ($Q_m < Q_c$)
- **Causes allocative inefficiency** — $P > MC$, meaning the value consumers place on extra units exceeds the cost of producing them, but the monopolist restricts output to keep prices high
- **Creates a deadweight welfare loss** — the triangle between the demand curve and MC, between Q_m and Q_c . This represents welfare that is neither consumer surplus nor producer surplus — it is simply lost.
- **Transfers surplus from consumers to producers** — consumer surplus shrinks; producer surplus (profit) grows

Arguments FOR Monopoly

- **Dynamic efficiency (Schumpeter's argument):** Supernormal profits provide the funding and incentive for research and development (R&D) and innovation. Joseph Schumpeter argued that monopolies drive "creative destruction" — the process by which innovation creates new products that render old ones obsolete. Without the prospect of monopoly profits, firms would have less incentive to innovate.
- **Natural monopoly:** In industries with very high fixed costs and large economies of scale (water, electricity grid, railways), having one firm is more efficient than having many. Duplication of infrastructure would be wasteful. These are best regulated rather than broken up.
- **Economies of scale:** A large monopolist may have lower average costs than many small firms. If regulated, these cost savings can be passed on to consumers as lower prices.
- **Cross-subsidisation:** A profitable monopolist can use profits from one area to fund socially valuable but unprofitable services. Royal Mail uses profits from urban areas to subsidise deliveries to remote rural areas.
- **International competitiveness:** Large domestic firms may be needed to compete against foreign giants on the global stage.

Arguments AGAINST Monopoly

- **Higher prices and lower output:** The fundamental welfare loss — consumers pay more and get less than they would in a competitive market.
- **Allocative inefficiency:** $P > MC$. Society's resources are not allocated according to consumer preferences. Too few resources go to the monopolised good.

- **Productive inefficiency:** Without competitive pressure, the monopolist has less incentive to produce at minimum average cost.
- **X-inefficiency:** Organisational slack — managers may allow costs to rise because there is no competitive threat to survival. Leibenstein's concept of X-inefficiency.
- **Inequality:** Monopoly profits are concentrated among shareholders (typically wealthier individuals), worsening the distribution of income and wealth.
- **Rent-seeking:** Monopolists may spend resources on lobbying, legal battles, and political influence to protect their monopoly position — these activities produce nothing of value for consumers.
- **Reduced consumer choice:** With only one supplier, consumers have no alternatives.

Real-World Example: Google's Search Monopoly

Google holds approximately 90% of the global search engine market. In 2017, the European Commission fined Google €2.4 billion for abusing its dominance by systematically favouring its own comparison shopping service in search results. Critics argue Google uses its monopoly in search to extend dominance into other areas (advertising, maps, email). Defenders argue Google's dominance reflects a superior product and that the search market remains contestable — users can switch to Bing or DuckDuckGo at zero cost. This case illustrates the tension between monopoly power and innovation.

Real-World Example: Pharmaceutical Patents

When Pfizer held the patent for Viagra (1998-2013), it could charge around £10 per pill. After the patent expired and generic competitors entered, the price fell to under £1 per pill. This illustrates both the case for and against monopoly: the patent gave Pfizer the incentive and funding to develop the drug in the first place (dynamic efficiency), but once granted, it charged monopoly prices that restricted access (allocative inefficiency). The patent system is society's compromise — granting temporary monopoly to incentivise innovation.

Natural Monopoly

A **natural monopoly** exists when the long-run average cost curve is still falling over the entire range of market demand. One firm can supply the whole market at a lower average cost than two

or more firms could. The MES is larger than or equal to total market demand.

Natural monopolies typically have:

- Very high fixed costs (infrastructure: pipes, cables, tracks)
- Very low marginal costs per additional user
- Duplication of infrastructure would be hugely wasteful

Examples: Water supply, electricity transmission, gas distribution networks, railways. It would be absurd to have two competing sets of water pipes or railway tracks running to every house.

Natural monopolies are typically **regulated** rather than broken up. Regulators (like Ofwat for water, Ofgem for energy) set price caps to prevent the monopolist from exploiting consumers while allowing the firm to cover costs and earn a reasonable return.

Oligopoly

Oligopoly is a market structure dominated by a small number of large firms, each with significant market share. The defining feature is **interdependence** — each firm must consider the likely reactions of its rivals when making pricing, output, and marketing decisions.

Characteristics of Oligopoly

- **High concentration ratio:** The top 3–5 firms hold a large share of the market. Measured by the *n-firm concentration ratio* (e.g., the 4-firm concentration ratio for UK supermarkets is about 67% — Tesco, Sainsbury's, Asda, and Morrisons).
- **High barriers to entry:** Economies of scale, brand loyalty, sunk costs, and patents protect incumbent firms.
- **Interdependence:** The key feature. Each firm's profit depends not just on its own actions but on the reactions of rivals. This makes decision-making strategic rather than independent.
- **Products may be differentiated or homogeneous:** Differentiated (cars, smartphones, supermarkets) or homogeneous (oil, steel, cement).
- **Non-price competition is common:** Advertising, branding, loyalty schemes, quality, innovation, customer service — because price wars can be destructive.

- **Price rigidity / "sticky prices"**: Oligopolists tend to keep prices stable even when costs change moderately. This is explained by the kinked demand curve model.

Real-World Example: UK Supermarkets

The UK grocery market is a classic oligopoly. Tesco (~27%), Sainsbury's (~15%), Asda (~14%), and Morrisons (~9%) collectively hold about 65% of the market. Each firm carefully monitors rivals' prices and promotions. When Asda launched a price-matching campaign, Tesco responded within weeks with its own "Aldi Price Match." When Aldi and Lidl (disruptors) grew rapidly, all the big four responded with budget ranges and price cuts. This interdependent behaviour — where each firm's strategy is a reaction to rivals — is the hallmark of oligopoly.

The Kinked Demand Curve Model

This model (developed by Paul Sweezy, 1939) explains why prices in oligopolistic markets tend to be **rigid/sticky** — they don't change much even when costs change.

The key assumption about how rivals react:

- If a firm **raises its price**, rivals will **NOT follow** — the firm loses customers to cheaper rivals. Demand is **elastic** above the current price.
- If a firm **cuts its price**, rivals **WILL match** the cut — nobody gains market share. Demand is **inelastic** below the current price.

This creates a "kink" in the demand curve at the current price, and a **vertical gap (discontinuity) in the MR curve**. As long as MC passes through this gap, the profit-maximising price and output remain unchanged — explaining sticky prices.

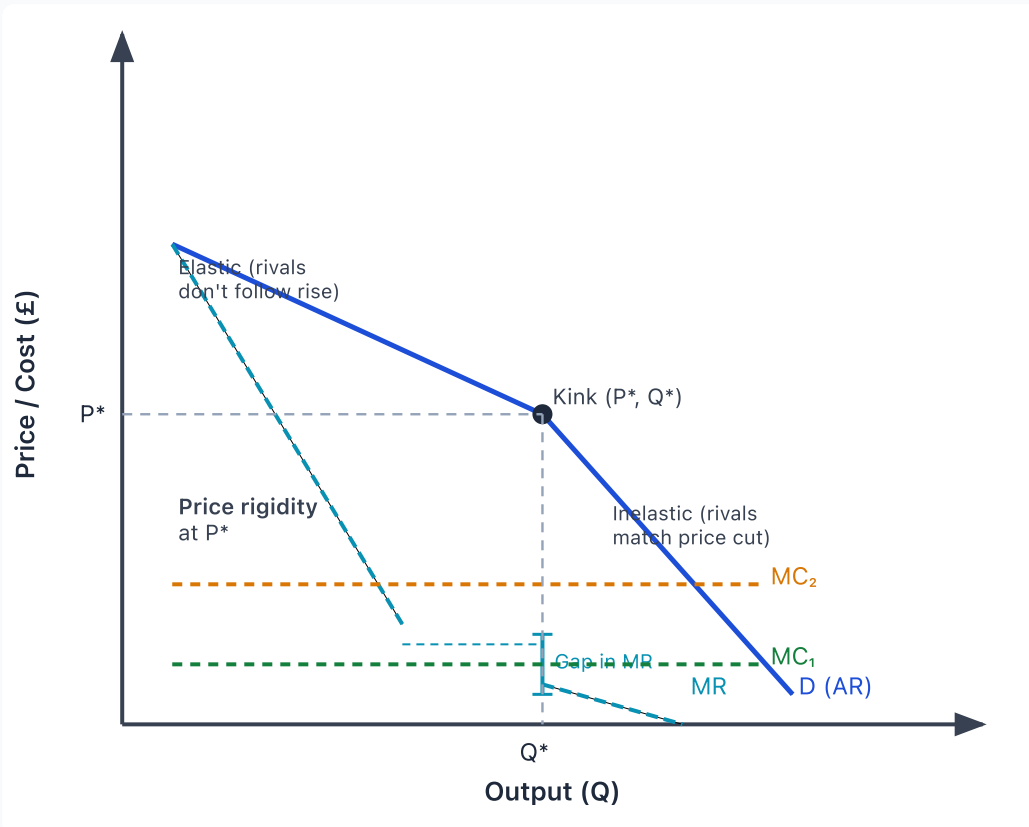


Figure 4: The kinked demand curve. MC can shift from MC_1 to MC_2 without changing the profit-maximising price P^* or output Q^* — explaining sticky prices in oligopoly.

Evaluation of the Kinked Demand Curve:

- **Strength:** It explains the observed real-world phenomenon of price rigidity in oligopolistic markets.
- **Weakness:** It does not explain *how* the current price was set in the first place — it only explains why it stays there.
- **Weakness:** The assumptions about rival reactions may not always hold. Sometimes rivals do follow price increases (tacit collusion) or don't follow price cuts.

Game Theory and the Prisoner's Dilemma

Game theory analyses strategic decision-making where the outcome for each player depends

on the choices of other players. It is the primary tool for modelling oligopolistic behaviour.

The **Prisoner's Dilemma** demonstrates why oligopolists often fail to cooperate, even when cooperation would benefit them all:

	Firm B	
	Low Price	High Price
Firm A: Low Price	A: £2m, B: £2m	A: £5m, B: £1m
Firm A: High Price	A: £1m, B: £5m	A: £4m, B: £4m

Analysis:

- If both set high prices (cooperate), both earn £4m each — the best joint outcome.
- But each firm has a **dominant strategy** to set a low price. Whatever Firm B does, Firm A is always better off choosing low price (£2m > £1m if B goes low; £5m > £4m if B goes high).
- The **Nash equilibrium** is (Low Price, Low Price) — both earn £2m. This is worse for both than cooperating, but neither has an incentive to change strategy unilaterally.
- This explains why price wars happen and why cartels are unstable — there's always an incentive to "cheat" on agreements.

Real-World Example: UK Supermarket Price Wars

The UK supermarket industry repeatedly demonstrates the Prisoner's Dilemma. In theory, all supermarkets would be more profitable if they maintained higher prices. But each has an incentive to undercut rivals to gain market share. The entry of Aldi and Lidl intensified this — the big four responded with price cuts, eroding everyone's margins. Tesco's operating margin fell from about 6% to under 3% between 2012 and 2018. The Nash equilibrium is lower prices and lower profits for everyone — exactly as game theory predicts.

Collusion

Collusion occurs when firms cooperate rather than compete, typically to raise prices and maximise joint profits. It can be overt (explicit agreement) or tacit (implicit coordination without a formal agreement).

Overt (Formal) Collusion — Cartels

A **cartel** is a formal agreement between firms to fix prices, limit output, or share markets. Cartels are **illegal** in the UK and EU under competition law. However, international cartels can be harder to prosecute.

Real-World Example: OPEC

OPEC (Organisation of the Petroleum Exporting Countries) is the most famous cartel. Its 11 member countries (after Angola left in 2024 and the UAE in May 2026) coordinate oil production quotas to influence global oil prices. When OPEC cuts production, supply falls and prices rise. However, OPEC faces the classic cartel instability problem: each member has an incentive to secretly exceed their quota (cheat) to earn more revenue at the higher price. This is why OPEC's discipline frequently breaks down and oil prices are volatile.

Tacit (Informal) Collusion

Firms coordinate behaviour **without any explicit agreement**:

- **Price leadership:** One firm (the market leader) sets a price and others follow. No formal agreement, but the outcome is the same as collusion.
- **Parallel pricing:** Firms independently set similar prices, watching each other's behaviour closely.

Factors Affecting the Likelihood of Collusion

Favours Collusion	Makes Collusion Difficult
Few firms (easier to coordinate)	Many firms (harder to monitor)
Similar cost structures	Different costs (disagreement on price)
Homogeneous products	Differentiated products

Stable, predictable demand	Volatile, changing demand
High barriers to entry	Low barriers (new entrants undercut)
Repeated interaction (long-term relationships)	One-off transactions
Weak competition authorities	Active regulators (CMA, EU Commission)

Non-Price Competition in Oligopoly

Because price wars are mutually destructive, oligopolists often compete on **non-price factors**:

- **Advertising and branding:** Creating brand loyalty to reduce PED and make demand more inelastic
- **Product innovation:** Launching new or improved products (Apple's annual iPhone updates)
- **Quality and features:** Competing on product quality, design, or included features
- **Loyalty programmes:** Tesco Clubcard, Nectar cards — increase switching costs for consumers
- **Customer service:** Free delivery, extended warranties, after-sales support
- **Location and convenience:** More stores, longer opening hours

Monopolistic Competition

Monopolistic competition is a market structure with *many* firms selling *differentiated* products, with *low barriers to entry and exit*. Each firm has a small degree of market power (its own downward-sloping demand curve) due to product differentiation, but faces strong competition from close substitutes.

Key Characteristics

- **Many buyers and sellers:** Many firms, but each is small relative to the market
- **Product differentiation:** Each firm's product is slightly different — through branding, quality, location, service, or design. This gives each firm a tiny "monopoly" over its specific version
- **Low barriers to entry and exit:** New firms can enter and existing firms can leave relatively easily

- **Some price-setting power:** Because products are differentiated, firms face a downward-sloping demand curve — but the degree of market power is limited by close substitutes
- **Non-price competition:** Firms compete heavily on advertising, branding, product quality, and service

Examples: Restaurants, coffee shops, hairdressers, clothing brands, independent pubs, local garages, small hotels.

Short Run: Supernormal Profit Possible

In the short run, a monopolistically competitive firm can earn supernormal profit (diagram similar to monopoly — $MC = MR$, with $P > AC$).

Long Run: Normal Profit Only

Because barriers to entry are low:

1. Supernormal profits attract new firms into the market
2. New firms take some customers away — the existing firm's demand curve shifts **left**
3. This continues until the demand curve is just **tangent** to the AC curve
4. At the tangency point: $P = AC \rightarrow$ normal profit only
5. But: $P > MC$ (allocative inefficiency) and not at minimum AC (productive inefficiency)

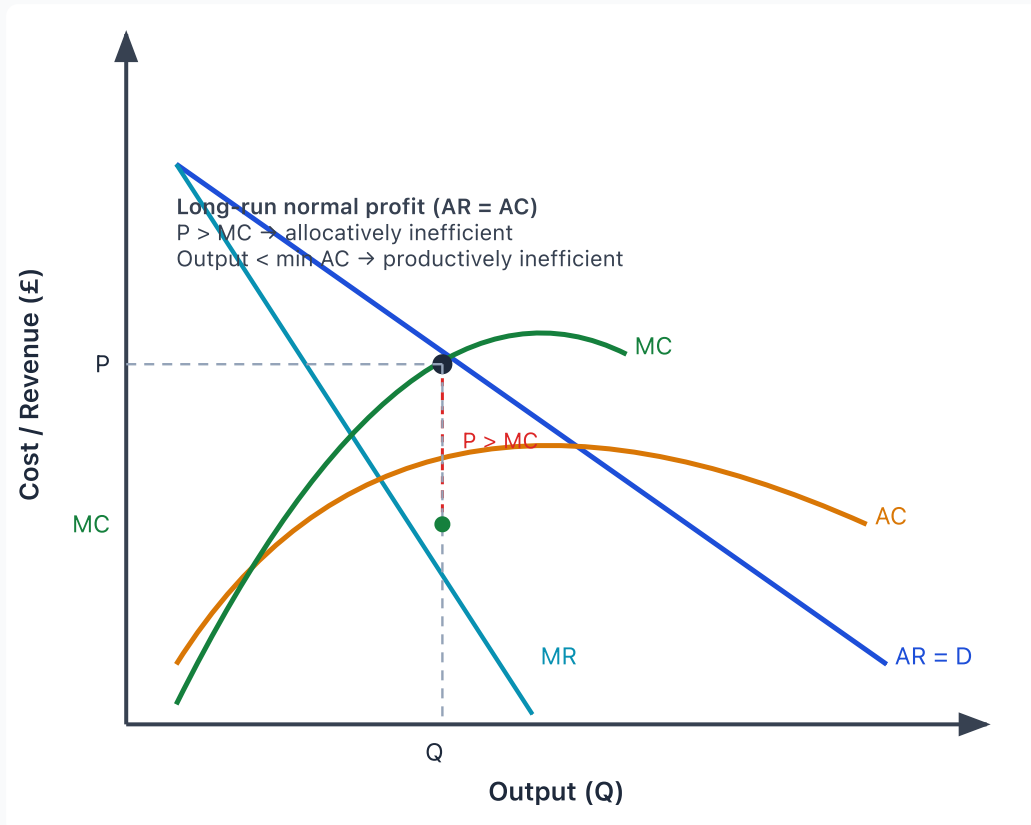


Figure 5: Long-run equilibrium in monopolistic competition. The demand curve is tangent to AC — normal profit only. But the firm is not allocatively or productively efficient.

Evaluation of Monopolistic Competition:

- **Inefficient:** $P > MC$ (allocative inefficiency) and not at minimum AC (productive inefficiency) — resources are wasted compared to perfect competition.
- **But:** Product differentiation provides **consumer choice and variety**. Consumers value having different types of restaurants, coffee shops, and clothing brands. The "cost" of inefficiency is the "benefit" of diversity.
- Some economists argue the welfare loss from monopolistic competition is small because firms have limited market power (many close substitutes).

Efficiency Across Market Structures

Allocative efficiency: $P = MC$ — resources are allocated according to consumer preferences. The value consumers place on the last unit equals its cost of production.

Productive efficiency: Production at the minimum point of AC — output is produced at the lowest possible cost per unit.

Dynamic efficiency: Firms invest in R&D, innovate, and improve products and processes over time. Often requires supernormal profits to fund investment.

X-efficiency: Firms operate without organisational slack — no wasted resources inside the firm. Driven by competitive pressure.

Efficiency	Perfect Competition	Monopolistic Competition	Oligopoly	Monopoly
Allocative ($P=MC$)	Yes (SR and LR)	No ($P > MC$)	Unlikely ($P > MC$)	No ($P > MC$)
Productive (min AC)	Yes (LR)	No (not at min AC)	Possible	Unlikely
Dynamic	Unlikely (no profits for R&D)	Limited (some differentiation)	Likely (profits + rivalry drive innovation)	Possible (profits fund R&D — Schumpeter)
X-efficiency	Yes (strong competitive pressure)	Mostly yes	Possible	Unlikely (no competitive pressure)

Exam Tip: There is a key trade-off between **static efficiency** (allocative + productive — perfect competition wins) and **dynamic efficiency** (innovation — monopoly/oligopoly may win). Use this as a central evaluation point when comparing market structures. The "best" structure depends on which type of efficiency you value most.

Contestable Markets

A **contestable market** (William Baumol, 1982) is one where there is freedom of entry and exit and where **sunk costs are zero or very low**. Even if the market is concentrated (few firms), the *threat* of potential entry disciplines existing firms to behave as if the market were competitive.

Conditions for Contestability

- **No or very low sunk costs:** The critical condition. Sunk costs are costs that cannot be recovered if the firm exits (e.g., specialised machinery, advertising). If sunk costs are zero, a firm can enter, take profits, and exit without loss — "hit-and-run" entry.
- **Freedom of entry and exit:** No legal or regulatory barriers preventing new firms from entering.
- **Access to the same technology:** Potential entrants can match incumbents' efficiency.
- **Perfect information:** Potential entrants know about profit opportunities.

Implications of Contestability

- Even a monopolist may be forced to set **competitive prices** if the threat of entry is credible.
- Firms may engage in **limit pricing** — deliberately setting prices low enough to make entry unprofitable, even though they could charge more.
- Supernormal profits are constrained by the *threat* of competition, even without actual competition.
- For government policy: reducing barriers to entry (deregulation, reducing sunk costs) may be more effective than breaking up monopolies.

Real-World Example: Airline Industry

The airline industry has some contestable features: aircraft can be leased (reducing sunk costs), and budget airlines like Ryanair and EasyJet have entered routes previously dominated by legacy carriers. However, perfect contestability is undermined by airport slot restrictions (Heathrow has almost no available slots), brand loyalty, frequent-flyer programmes, and the high cost of establishing new routes. So while more contestable than, say, water supply, the airline market is far from perfectly contestable.

Evaluation: Contestability in Practice

- Contestability is a powerful evaluation tool — always consider whether a concentrated market faces the *threat* of entry, not just actual competition.
- However, **perfect contestability is rare** in practice. Most markets have some sunk costs, brand loyalty, and information asymmetries.
- The theory suggests policy should focus on **reducing barriers to entry** (deregulation, opening markets, reducing switching costs) rather than simply breaking up large firms.
- The rise of digital markets has increased contestability in some sectors (online retail, streaming) but created new barriers in others (network effects, data advantages).

Price Discrimination

Price discrimination occurs when a firm charges different prices to different consumers for the same product, where the price difference is not justified by a difference in cost. It requires: (1) market power, (2) the ability to identify different groups with different PED, and (3) the ability to prevent resale between groups.

Types of Price Discrimination

Degree	Description	Example
First degree (perfect)	Each consumer is charged the maximum they are willing to pay. All consumer surplus is captured by the producer.	Haggling in a market, personalised online pricing (algorithmic), auction outcomes
Second degree	Different prices charged for different quantities purchased (bulk discounts).	"Buy 2 get 1 free," tiered pricing for electricity, multi-buy deals
Third degree	Different prices charged to different identifiable groups (segmented by age, time, location, etc.).	Student discounts, peak/off-peak rail fares, cheaper cinema tickets for children/OAPs

Conditions Required for Price Discrimination

1. **Market power:** The firm must be able to set prices (not a price taker)
2. **Ability to segment the market:** The firm must identify groups with different price elasticities of demand
3. **Prevention of resale (arbitrage):** Consumers in the cheap segment cannot resell to those in the expensive segment

Real-World Example: Train Tickets

UK train companies practise extensive third-degree price discrimination. A peak-time London to Manchester ticket might cost £180, while an off-peak advance ticket for the same journey costs £30. The product is identical — the same seat on a similar train. Business travellers (who must travel at peak times) have inelastic demand (their employer pays, they can't be flexible). Leisure travellers have elastic demand (they can choose when to travel, or drive instead). By charging different prices, the train company captures more consumer surplus and increases total revenue.

Evaluation of Price Discrimination

Arguments FOR	Arguments AGAINST
Some consumers pay less than they would under a single price (e.g., students, off-peak travellers)	Some consumers pay more than the single price — reduction in their consumer surplus
Firms earn more revenue, which can fund investment and innovation	Allocative inefficiency may worsen (different P/MC ratios in different segments)
May allow services to operate that wouldn't exist at a single price (e.g., off-peak train services)	Can be seen as unfair — why should one person pay more than another for the same product?
Cross-subsidisation: profits from one group fund cheaper access for another	First-degree discrimination eliminates all consumer surplus

Evaluation: Which Market Structure is Best?

Key Evaluation Points for Essays:

- **No single structure is universally "best"** — the answer depends on which criteria you value (static efficiency, dynamic efficiency, consumer welfare, innovation, choice).
- **Perfect competition** delivers static efficiency ($P = MC$, min AC) but may lack dynamic efficiency — firms have no supernormal profits to fund R&D.
- **Monopoly** is statically inefficient ($P > MC$, deadweight loss) but Schumpeter argued it can be dynamically efficient — the prospect of monopoly profits drives innovation. Patents exist for this reason.
- **Oligopoly** may deliver a balance — firms have both the profits to fund R&D and the competitive rivalry to drive innovation. Many of the world's most innovative industries are oligopolies (tech, pharmaceuticals, automotive).
- **Contestability matters more than structure** — a monopoly in a highly contestable market may behave more competitively than a cartel-like oligopoly in a non-contestable market.
- **Context is everything:** The nature of the good, the regulatory environment, technology, and consumer preferences all determine which structure produces the best outcomes. Natural monopolies should be regulated, not broken up. Competitive markets may be best for simple consumer goods.